Alberto Testoni, Raffaella Bernardi, and Azzurra Ruggeri. 2023. The efficiency of question-asking strategies in a real-world visual search task. *Cognitive Science*, 47(12).

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. 2023. Llama 2: Open foundation and fine-tuned chat models. *Preprint*, arXiv:2307.09288.

Thinh Hung Truong, Timothy Baldwin, Karin Verspoor, and Trevor Cohn. 2023. Language models are not naysayers: an analysis of language models on negation benchmarks. In *Proceedings of the 12th Joint Conference on Lexical and Computational Semantics (*SEM 2023)*, pages 101–114, Toronto, Canada. Association for Computational Linguistics.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander Rush. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. 2022. Star: Bootstrapping reasoning with reasoning. In *Advances in Neural Information Processing Systems*, volume 35, pages 15476–15488. Curran Associates, Inc.

Yizhe Zhang, Jiarui Lu, and Navdeep Jaitly. 2024. Probing the multi-turn planning capabilities of LLMs via 20 question games. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1495–1516, Bangkok, Thailand. Association for Computational Linguistics.

A Methodology

A.1 Successful Condition

A dialogue is *successful* when the Questioner guesses the target entity. We have implemented this condition with the following two cases: (a) the Questioner explicitly refers to the target entity without naming other candidate entities or (b) the Questioner explicitly refers to the target entity along with other candidates, and the target is named more times than each of the other entity. This second condition recognizes when the Questioner is explicitly reasoning about the target before guessing (e.g., target: 'cherry'. Questioner: 'So, the target is one of the remaining fruits, which are cherry and grapefruit. Let me make a guess. Is the target... cherry?'). The dialogue is considered *unsuccessful* if the Questioner does not meet the conclusive conditions above within 20 questions.

A.2 Expected Information Gain

In this setting, each question aims to rule out as many candidates as possible from the set $\Omega = \{\omega_1, \omega_2, ..., \omega_n\}$. The most informative question rules out half of the possible candidates, whereas a question ruling out none or just one candidate has low informativeness. Questions' informativeness has been quantified in prior works through the Expected Information Gain.

Expected Information Gain (EIG) measures the contribute of each question to reduce uncertainty towards the solution ω. The level of uncertainty is measured through the entropy H, higher entropy means higher uncertainty. Consequently, EIG is computed subtracting from the initial state of entropy (H_{prior}) the expected entropy after asking the question ($H_{posterior}$).

$$EIG = H_{prior} - H_{posterior} \tag{1}$$

At the beginning of each game, we assume a uniform prior distribution over all the possible candidates $\omega_1, \omega_2, ..., \omega_n$. From the Shannon entropy (Shannon, 1948), this initial state H_{prior} in our setting is equal to:

$$\begin{aligned} H_{prior} &= -\sum_{i=1}^{n} p(w_i)\, log_2\, p(w_i) \\ &= -n\,\frac{1}{n}\, log_2\,\frac{1}{n} \;=\; log_2\, n \end{aligned} \tag{2}$$

A yes/no question divides the current candidates in Ω into two disjoint subsets: $\Omega_{yes} \cup \Omega_{no}$. In

the subset Ω_{yes} there are the candidates in Ω that are consistent with a possible positive answer, Ω_{no} with a negative one. Answering the question, the Answerer determines which subset becomes the new set of candidates under consideration for the next turn (the candidates in the other subset are ruled out). The posterior entropy is computed as the sum of the entropies of both the yes and no subsets, weighted by their probability. $H_{posterior}$ is computed as:

$$H_{posterior} = p_{yes}\, H_{yes} \; + \; p_{no}\, H_{no} \qquad (3)$$

This computation is based on the expected answers to the question for each candidate in the current set. In other words, the Questioner has an expectation of which candidates will be in Ω_{yes} and Ω_{no} before receiving the answer by the Answerer. With these subsets, the probabilities p_{yes} and p_{no} refer to the probabilities of receiving a positive or negative answer. The entropies H_{yes} and H_{no} measure uncertainty in the two subsets computed as in eq. 2. Intuitively, the higher the similarity between the two subsets in terms of size (i.e., candidates per subset) the higher the EIG value. The most informative yes/no questions divide Ω into two subsets Ω_{yes} and Ω_{no} of the same size, resulting in $EIG = 1$.

A.3 Setting

A.3.1 Sampling

Our approach comprises the three steps described in Fig. 3. During sampling, different questions are sampled from the Questioner. To compute EIG, the Annotator annotates the questions with yes/no for each candidate in the candidate set (Ω). Once an 'optimal' question is reached ($EIG = 1$), the optimal and suboptimal questions are stored in the training datasets. All the prompts for the roles are reported below. The Annotator has the same prompt as the Answerer, taking as a target each candidate in the candidate set.

Questioner

System Prompt: *You are playing a game, make only one yes/no question at turn to identify the target from the List of candidates. If there are 1 or 2 candidates remaining make the guess.*
User Prompt: *List of candidates:* `CANDIDATES`.

Answerer/Annotator

System Prompt: *You are playing the 20-Questions game, you will be asked one Question about the Target element. Answer only 'yes' or 'no' to the Question depending on your Target element.*
User Prompt: *Target element:* `TARGET`, *Question:* `QUESTION`

To yield diverse outputs during sampling, the decoding strategy for the Questioner involves a relatively high temperature (1.0) and top-k sampling (50). To achieve more deterministic responses the Answerer/Annotator has a low temperature of 0.1.

This process produces a Fine-Tuning and DPO dataset. The Fine-Tuning dataset consists of all the completed dialogues of optimal questions only. The DPO dataset, instead, comprises pairs of 'optimal' questions ($EIG = 1.0$) with suboptimal questions ($EIG < 0.8$).

A.3.2 Training

The Fine-Tuning and DPO datasets are employed to train LLAMA 2-CHAT (7B) with LoRA adapters (Hu et al., 2022). Our main focus is on the approach rather than the training strategy, thus we did not perform extensive hyperparameter tuning. We do not exclude that additional experiments on hyperparameters could lead to better results for our approach. For both Fine-Tuning and DPO, the LoRA adapters are applied on all the modules, with an $r = 128$ and the alpha value $= 32$. We perform Fine-Tuning and DPO with AdamW as the optimizer, training for one epoch and with a batch size $= 8$. For both Fine-Tuning and DPO, a learning rate of $2e-5$ has been observed to produce consistent results.

A.3.3 Testing

At test time, we compare the LLAMA 2-CHAT (7B) zero-shot with its Fine-Tuned and DPO alternatives.

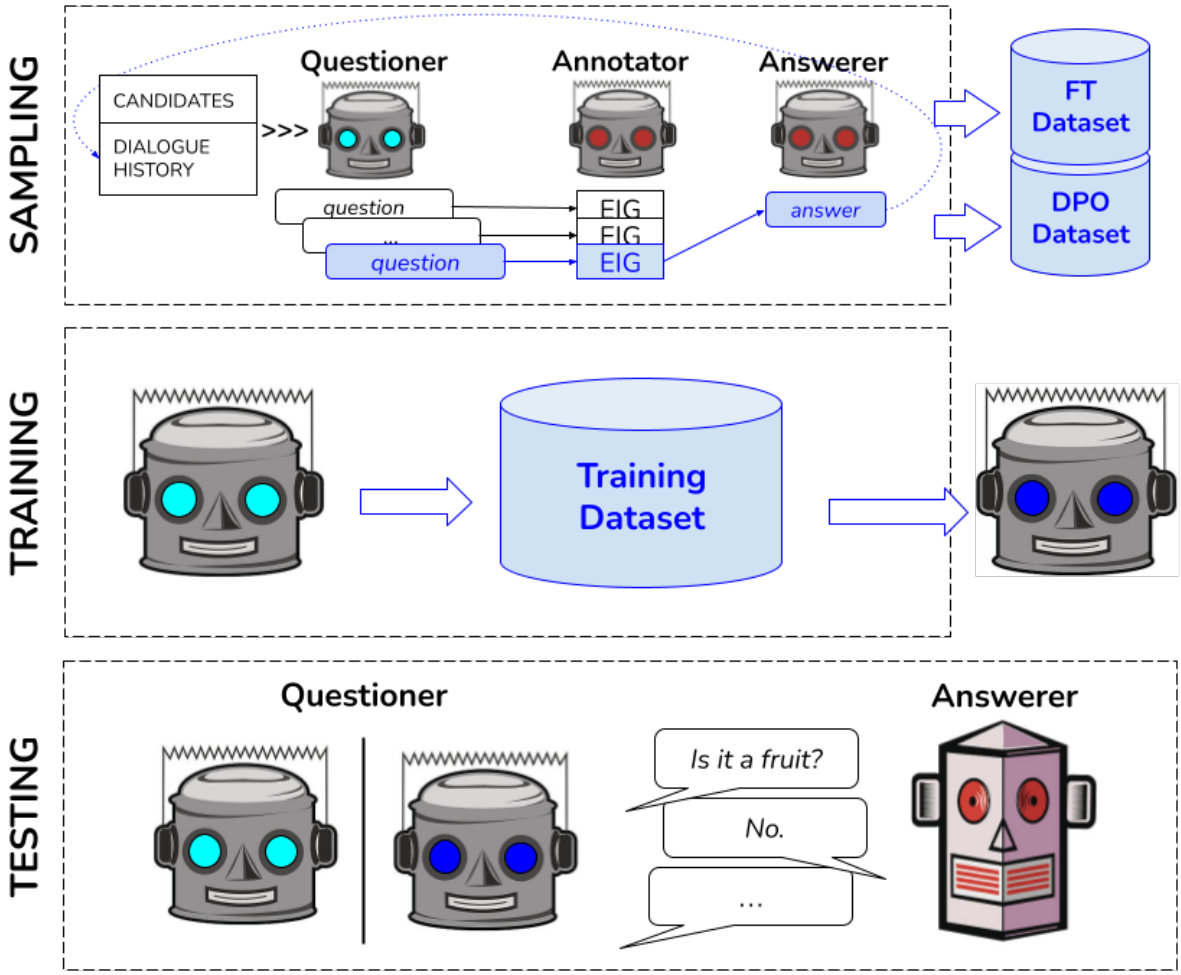

Figure 3: During **sampling**, LLAMA 2-CHAT (7B) plays the roles of the Questioner, the Annotator, and the Answerer. Questions are sampled from the Questioner and then evaluated by the Annotator. Once an optimal question is reached, the Answerer answers it. The optimal question and its answer are appended to the dialogue history. In this way, optimal questions are sampled not only for the first turn but also in follow-up turns. In **training**, the Questioner is trained with FT and DPO datasets. In **testing**, the zero-shot 🤖 and trained 🤖 Questioners play the 20 Questions Game with an external model as Answerer.